

Practice Quiz 07D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

Construct a function whose value at a point differs from its limit.

1. Construct a function $f : \mathbf{R} \rightarrow \mathbf{R}$ such that

$$f(0) < 0 \quad \text{and} \quad \lim_{x \rightarrow 0} f(x) = 1.$$

Give a formula defining f for every real input, and prove the limit directly from the epsilon-delta definition.

Solution Define

$$f(x) = \begin{cases} -1, & x = 0, \\ 1, & x \neq 0. \end{cases}$$

This defines a real value for every real input, and $f(0) = -1 < 0$.

To prove the limit, let $\varepsilon > 0$ be arbitrary and choose $\delta = 1 > 0$. If $x \in \mathbf{R}$ satisfies $0 < |x| < \delta$, then $x \neq 0$, so $f(x) = 1$. Consequently

$$|f(x) - 1| = 0 < \varepsilon.$$

This verifies the definition and proves $\lim_{x \rightarrow 0} f(x) = 1$. The condition $0 < |x|$ excludes the input at which we assigned -1 . ■